

An Improved, Durable Bitter-Type Electromagnet for Cold Atom Experiments

Henry Ando,¹ Leon Gold,¹ Geyue Cai,¹ Shaozhen Yang,¹ Mohit Verma,¹ David DeMille,² and Cheng Chin³

¹*The James Franck Institute and Department of Physics, The University of Chicago, Chicago, Illinois 60637, USA*

²*William H. Miller III Department of Physics and Astronomy, Johns Hopkins University, Baltimore, Maryland 21218, USA*

³*The James Franck Institute, Enrico Fermi Institute, and Department of Physics, The University of Chicago, Chicago, Illinois 60637, USA*

(*cchin@uchicago.edu)

(Dated: 11 August 2026)

We present the design, construction, and characterization of an upgraded, high-reliability Bitter-type electromagnet optimized for generating large, continuous-duty magnetic fields in cold atom experiments. Building upon the design reported in Sabulsky et al.¹ and our experience in the past 12 years, this iteration (1) improves water distribution to the coil, minimizing possible leaks, and (2) improves mechanical clamping forces on the electrical current pathways to eliminate heating due to contact conductance degradation. At a continuous operating current of 350 A and a cooling water flow rate of 3.8 L/min, the coil body exhibits a steady-state temperature rise of only 6 °C, and the heating of other elements is mitigated almost entirely. Overall, the design modifications presented here drastically reduce peak thermal resistance and operational risks, providing an easily maintainable and highly stable magnetic field source for demanding ultracold experiment applications.

I. INTRODUCTION

Modern cold atom experiments require the generation of large and highly stable magnetic fields, rapid field control, and exceptional thermal stability. With many experiments requiring fields exceeding 1000 G, a key technical challenge is engineering field generators that can handle high power dissipation without introducing thermal drifts or structural risks to the surrounding vacuum apparatus.

Beyond the most standard approach to generate magnetic fields on wound, hollow-core copper wire coils, we reported on the construction of a Bitter coil optimized for cold atom apparatuses¹. Many related designs have been developed subsequently^{2–8}.

Bitter coils⁹ offer key advantages over wound hollow-core wire coils: (1) Parallel cooling. The water cooling channels run in parallel rather than in series with the electrical current, resulting in a more uniform temperature distribution across the coil body. (2) Low hydrodynamic resistance. The internal fluid paths offer significant less resistance to water flow, eliminating the need for high-pressure pumps which can generate mechanical instability due to turbulence. (3) Higher magnetic field at lower inductance. Bitter coils are made by connecting copper disks, which offers higher copper packing ratio, radial current distribution that favors magnetic field generation and lower electrical inductance than hollow-core magnetic wires. (4) Standard manufacturing processes. A Bitter coil can be made using 3D printing, laser and waterjet cutting, and standard machining processes, all of which feature high precision and repeatability. The coils can be assembled and reassembled without epoxy.

After over a decade of continuous operational use, we have identified two primary long-term vulnerabilities in the original architecture¹. First, the electrical contact resistance of the bolts that carry current into and out of the primary coil

stack tends to creep upward over time due to thermal cycling and surface oxidation. This causes extreme localized heating that is difficult to service due to physical access limitations. Second, the traditionally machined plastic water-distribution manifold was prone to cracking and leaking over extended life cycles.

In this work, we present a comprehensive redesign that resolves these vulnerabilities by leveraging modern fabrication technologies, including water-jet cutting, laser cutting, and 3D printing. Our design modifications draw inspiration from recent developments in complex Bitter geometries³. Our updated current delivery system decouples mechanical clamping forces from the electrical pathways, replacing failure-prone internal threaded contacts with flat, easily accessible, and water-cooled terminal surfaces. Additionally, our new monolithic 3D printed water manifold reduces the total number of external fluid connections from eight down to two and entirely eliminates structural threads tapped directly into plastic. With these structural improvements, we reduce the maximum temperature increase to just 6 °C when producing a 1000 G field with a 350 A continuous current, in a design that we believe will be substantially more robust and maintainable than its predecessor.

II. CURRENT DELIVERY SYSTEM

The main electrical issues we've encountered with the original version of the coil have to do with the connections between the bus bars and the coil stack itself.

In the original design, a current-carrying screw carried current from the bottom layer of the coil to the outgoing bus bar. The full current then passes through the threads of the screw into a copper block. This electrical connection proved problematic over time, as thermal cycling and oxidation gradually

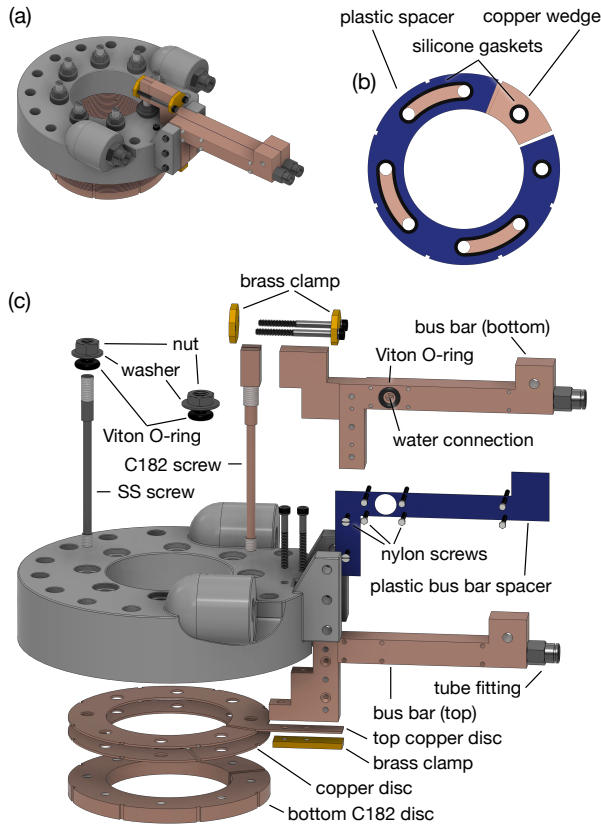


FIG. 1: Bitter coil design and components. (a) An assembled Bitter coil with water cooled bus bars. (b) A single layer copper disc is covered with a plastic insulating spacer to prevent it shorting to the next layer, silicone gaskets to seal the water into well defined cooling channels, and a thinner copper wedge to provide an electrical connection to the next layer. (c) The key parts of the new Bitter coil design are shown in an expanded view.

caused a runaway increase in contact resistance. Undoing this connection to clean and deoxidize coil surfaces is impossible because the connection in question also provides the clamping force that holds the coil together and the water in.

Another problematic connection was where current was carried from the incoming bus bar into the second current-carrying screw through the threads, then passed into the copper distribution block through the tip of the screw. Both of these connections also developed high contact resistances over time. Since the copper distribution block was recessed, our ability to clean and polish its surface to improve the contact resistance was minimal.

In the updated design, the direction of the compression bolts is reversed: they are installed from the water distribution block downward into blind-tapped holes in the bottom plate (Fig. 1). This enables us to largely separate the mechanical and electrical roles of the current-carrying screw. Now, a nut provides mechanical compression on the coil while the electrical connection is made by a flat surface clamped securely

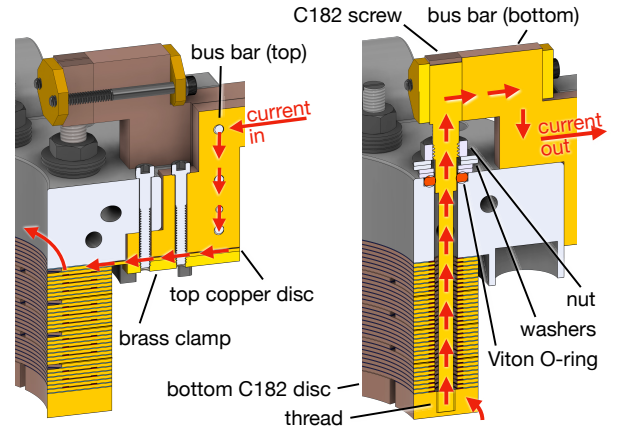


FIG. 2: Cross-section view of the current path into and out of the coil through the bus bars. (a) A tongue of copper extends out from the top layer of the coil and is clamped securely to the bus bar by a small brass clamping block from below. (b) The bottom copper 182 disc is electrically and mechanically connected to a copper 182 clamping screw through a blind tap. The copper 182 screw is electrically connected to the bus bar through a large flat surface that is securely clamped, and a separate nut with a Viton O-ring allows independent sealing and compression of coil stack.

onto the bus bar, see Fig. 2b. Thus, the bus bar can be removed and the connection cleaned, without breaking the water seal. Should the threaded connection between the current-carrying screw and the bottom plate need to be serviced, the blind taps allow the current-carrying screw to be removed through the top of the water distribution block without creating a leak at the bottom of the coil.

Meanwhile, the connection between the incoming bus bar and the top layer of the coil is now made directly, by adding a tongue of metal to the top layer, which is clamped to the bottom of the incoming bus bar, see Fig. 2a. The contact area here is much greater than before, and the bus bar can be removed and all relevant surfaces cleaned/deoxidized if need be, without creating any leaks.

Another general improvement in the new design is that all parts serving both electrical and mechanical purposes (which were previously made of brass) we now make out of the chromium-copper alloy copper 182, inspired by Siegel et al.³. C182 has twice the electrical conductivity of brass while also being stronger and more chemically similar to copper.

The bus bars also generate nontrivial amounts of heat. The updated design therefore features water-cooling for the bus bars, which we consider mandatory. In this design, the bus bars are cooled via internal machined water channels, and water is circulated from one bus bar to the other via a pass-through sealed by a Viton O-ring (Fig. 1). The two bus bars are held together via Nylon screws and separated by a plastic bus bar spacer to prevent shorting. This configuration allows both bus bars to be cooled with a single supply/return water line pair, which is convenient but not required. The bus bars pictured here are also longer than generally needed due to

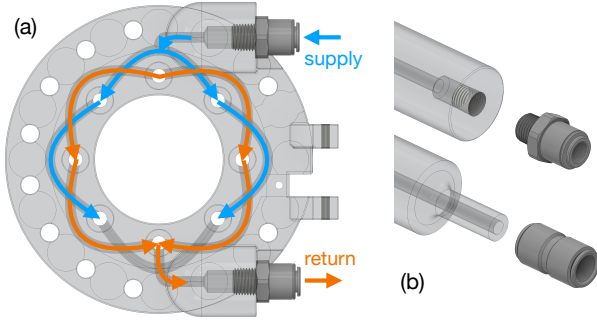


FIG. 3: (a) Flow in the internal channels of the 3D printed water distribution manifold. (b) Two ways to connect external water tubes to 3D printed water distribution manifolds: NPT threads with plastic male NPT fittings and plastic pipe thread sealant, or 3D printing external tube geometry for use with push-to-connect fittings.

space constraints of our specific system.

III. WATER DISTRIBUTION SYSTEM

The primary issue encountered with the water distribution system in the past was leaks. Leaks mainly occur when the plastic around an NPT-tapped pipe fitting cracks. Once a crack forms, little can be done to remedy the problem in the long term. Common plumbing wisdom advises avoiding plastic/metal pipe connections, especially when a metal male fitting is installed into plastic female threads. This is because plastic is much weaker than metal, especially under tension. Moreover, plastic and metal have different coefficients of thermal expansion, so with thermal cycling, the metal fittings expand and contract relative to their plastic sockets, gradually stressing the material until it eventually cracks.

The prior design was additionally prone to leaks due to its four water inlets and four outlets. These eight fluidic connections were necessary due to the geometry limitations of traditional machining, and all eight featured metal NPT fittings screwed into plastic female threads.

In the new design, we leverage the capabilities of 3D printing to address both of the above issues. 3D printing the water distribution block allows us to reduce the number of water inlets and outlets from four each to one each, by routing the water between the vertical channels through internal 3D printed water channels, see Fig. 3. This technique was also used by Siegel et al.³

We also report on two ways to avoid screwing metal male NPT fittings into female plastic threads, see Fig. 3b. One is simply by switching to plastic male NPT threads and using appropriate plastic pipe thread sealant paste. The other is by 3D printing an external 0.5" diameter tube, onto which a normal push-to-connect tube fitting can be installed without modification. This eliminates any tensile strain on the plastic piece. We have achieved secure, leak-free connections with both techniques.

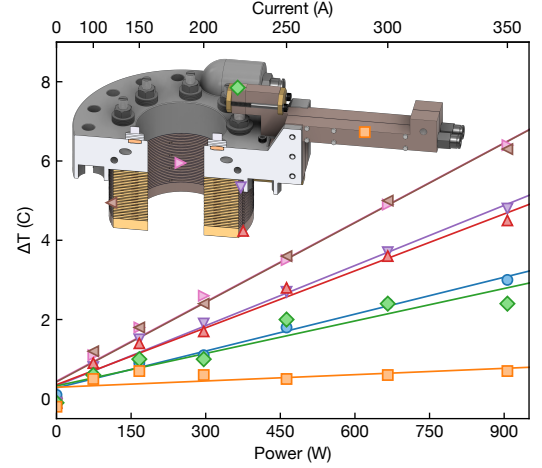


FIG. 4: Temperature increase ΔT relative to water supply in various parts of the coil as a function of dissipated power. The return water temperature is indicated by blue circles. All measurements are taken with 3.8 L/min of water flowing through the coil, at a pressure drop of 40 kPa. The supply water temp is approximately 21 °C. Applied current is plotted on the top x-axis for ease of comparison. Solid lines are a linear fit to the data.

IV. RESULTS

Several of the key improvements implemented here do not correspond to any particular metric, but rather to the overall durability and maintainability of the coil. The main performance metrics of merit can be broken down into electrical and temperature performance.

Electrically, the inductance of the new coil is measured to be 43 μH , comparable to the previous version and to the theoretical prediction. The overall resistance is measured to be 7.2 $\text{m}\Omega$, nearly 2 $\text{m}\Omega$ lower than the previous iteration, where much of the improvement comes from the redesigned bus bar contacts.

The temperature performance of the new coil is shown in Fig. 4. At 3.8 L/min of flow, the temperature increase of the hottest measured point on the coil is kept around 7 °C/kW, and the spread in temperatures across the coil body is less than 3 °C/kW. There is no appreciable heating of the bus bars now, whereas the un-cooled bus bars of the old coil would regularly reach $> 50^\circ\text{C}$ during normal performance at 300 A. This demonstrates the importance of water-cooling all the current-carrying elements of the coil.

V. CONCLUSION

We have described a state of the art, simple, and scalable design of a Bitter electromagnet design for cold-atom experiments alongside other potential uses in industry or quantum computing. During high field operation the coil exhibits stable resistances, temperatures, and magnetic fields, and outper-

forms many peer electromagnets, including the prior design published from this group.

The new coil provides improved thermal and electrical performance compared to its predecessor. Additionally, the new design is more robust, maintainable, and mass-producible. The new cooling system eliminates leaks, reduces tension, and has simplified plumbing, with only one inlet and one outlet.

Overall, this design details a low cost, state of the art, thermally stable, and machinable electromagnet to provide stable fields for cold atom experiments.

VI. ACKNOWLEDGEMENTS

H.A. would like to acknowledge helpful discussions with Daniel Barker. Krutik Patel gave valuable early advice on the modifications to the design. We would like to acknowledge the UChicago MRSEC for use of its facilities. The work in Cheng Chin's group was supported by the National Science Foundation (NSF) under Grant No. PHY-2409612 and by the Air Force Office of Scientific Research under

Award No. FA9550-21-1-0447. H. A. acknowledges support by the NSF Graduate Research Fellowship under Grant No. DGE 1746045. M.V. acknowledges support from the Natural Sciences and Engineering Research Council of Canada (NSERC).

VII. CITATIONS

- ¹D. O. Sabulsky, C. V. Parker, N. D. Gemelke, and C. Chin, *Review of Scientific Instruments* **84**, 104706 (2013).
- ²T. Luan, T. Zhou, X. Chen, and Z. Ma, *Review of Scientific Instruments* **85**, 024701 (2014).
- ³J. L. Siegel, D. S. Barker, J. A. Fedchak, J. Scherschligt, and S. Eckel, *Review of Scientific Instruments* **92**, 033201 (2021).
- ⁴E. G. Hataway, E. K. Falk, K. E. O'Neill, M. P. Berghof, and B. A. Olsen, 10.48550/arXiv.2607.02813, 2607.02813.
- ⁵Y. Long, F. Xiong, V. Gaire, C. Caligan, and C. V. Parker, *Physical Review A* **98**, 043626 (2018).
- ⁶K. Roux, B. Cilenti, V. Helson, H. Konishi, and J.-P. Brantut, *SciPost Physics* **6**, 048 (2019).
- ⁷R. Koirala and B. A. Olsen, **97**, 033205.
- ⁸N. McKay Parry, M. Baker, T. Neely, T. Carey, T. Bell, and H. Rubinsztein-Dunlop, *Review of Scientific Instruments* **85**, 086103 (2014).
- ⁹F. Bitter, *Review of Scientific Instruments* **7**, 482 (1936).